



LIMITING REAGENTS 2

- 1 What mass of calcium hydroxide is formed when 10.0 g of calcium oxide reacts with 10.0 g of water? $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

$$\text{Moles of CaO} = \frac{10.0}{56} = 0.179 \text{ mol}$$

$$\text{Moles of H}_2\text{O} = \frac{10.0}{18} = 0.556 \text{ mol}$$

0.179 mol of CaO reacts with 0.179 mol of H₂O, ∴ H₂O is in excess; CaO is limiting reagent

moles of Ca(OH)₂ formed = 0.179 mol

mass of Ca(OH)₂ = 74 x 0.179 = 13.2 g

- 2 What mass of magnesium bromide is formed when 1.00 g of magnesium reacts with 5.00 g of bromine? $\text{Mg} + \text{Br}_2 \rightarrow \text{MgBr}_2$

$$\text{Moles of Mg} = \frac{1.000}{24} = 0.0417 \text{ mol}$$

$$\text{Moles of Br}_2 = \frac{5.00}{160} = 0.03125 \text{ mol}$$

0.03125 mol of Mg reacts with 0.03125 mol of Br₂, ∴ Mg is in excess; Br₂ is limiting reagent

moles of MgBr₂ formed = 0.03125 mol

mass of MgBr₂ = 184 x 0.03125 = 5.75 g

- 3 What mass of copper is formed when 2.00 g of copper(II) oxide reacts with 1.00 g of hydrogen? $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

$$\text{Moles of CuO} = \frac{2.00}{79.5} = 0.0252 \text{ mol}$$

$$\text{Moles of H}_2 = \frac{1.00}{2} = 0.500 \text{ mol}$$

0.0252 mol of CuO reacts with 0.0252 mol of H₂, ∴ H₂ is in excess; CuO is limiting reagent

moles of Cu formed = 0.0252 mol

mass of Cu = 63.5 x 0.0252 = 1.60 g

- 4 What mass of sodium fluoride is formed when 2.30 g of sodium reacts with 2.85 g of fluorine? $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$

$$\text{Moles of Na} = \frac{2.30}{23} = 0.100 \text{ mol}$$

$$\text{Moles of F}_2 = \frac{2.85}{38} = 0.075 \text{ mol}$$

0.100 mol of Na reacts with 0.050 mol of F_2 , $\therefore \text{F}_2$ is in excess; Na is limiting reagent

moles of NaF formed = 0.100 mol

$$\text{mass of NaF} = 42 \times 0.100 = 4.20 \text{ g}$$

- 5 What mass of iron is formed when 8.00 g of iron(III) oxide reacts with 2.16 g of aluminium? $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$

$$\text{Moles of Fe}_2\text{O}_3 = \frac{8.00}{160} = 0.050 \text{ mol}$$

$$\text{Moles of Al} = \frac{2.16}{27} = 0.080 \text{ mol}$$

0.040 mol of Fe_2O_3 reacts with 0.080 mol of Al, $\therefore \text{Fe}_2\text{O}_3$ is in excess; Al is limiting reagent

moles of Fe formed = 0.080 mol

$$\text{mass of Fe} = 56 \times 0.080 = 4.48 \text{ g}$$

- 6 What mass of aluminium chloride is formed when 13.5 g of aluminium reacts with 42.6 g of chlorine? $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$

$$\text{Moles of Al} = \frac{13.5}{27} = 0.500 \text{ mol}$$

$$\text{Moles of Cl}_2 = \frac{42.6}{71} = 0.600 \text{ mol}$$

0.400 mol of Al reacts with 0.600 mol of Cl_2 , $\therefore \text{Al}$ is in excess; Cl_2 is limiting reagent

moles of AlCl_3 formed = 0.400 mol

$$\text{mass of AlCl}_3 = 133.5 \times 0.400 = 53.4 \text{ g}$$

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Can find mass of product		
Shows suitable working			Can identify limiting reagent			Gives units		
Can work out M_r			Can deduce moles of product					